

GSLs optimizer crashing with QAOA default parameter bounds

<https://github.com/qiskit-community/qiskit-algorithms/issues/94>

ROW 58

GSLs optimizer crashing with QAOA default parameter bounds #94

New issue

Closed

#95



dieidreiter opened on Oct 23, 2023

Environment

- Qiskit Algorithms version: 0.2.1
- Python version: 3.11.1
- Operating system: Windows 10

What is happening?

When using the `GSLs` optimizer in combination with the `QAOA` `minimum_eigsolver`, the optimization crashes. This seems to stem from the `GSLs` optimizer not being capable of dealing with parameter bounds of the form `[(None, None)]`.

How can we reproduce the issue?

The bug can be reproduced by trying to compute the minimum eigenvalue of a trivial hamiltonian:

```
from qiskit import Aer
from qiskit.primitives import BackendSampler
from qiskit.quantum_info import SparsePauliOp
from qiskit_algorithms.optimizers import GSLs
from qiskit_algorithms.minimum_eigsolvers import QAOA

backend = Aer.get_backend("aer_simulator")
sampler = BackendSampler(backend)

optimizer = GSLs()
qaoa = QAOA(optimizer=optimizer, sampler=sampler)

hamiltonian = SparsePauliOp(["ZI"])
result = qaoa.compute_minimum_eigenvalue(hamiltonian)
print(result)
```

This results in the following Traceback:

```
Traceback (most recent call last):
  File "C:\...\minimal_example.py", line 19, in <module>
    result = qaoa.compute_minimum_eigenvalue(hamiltonian)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\...\venv\Lib\site-packages\qiskit_algorithms\minimum_eigsolvers\sampling_vqe.py", line 218, in compute_minimum_eigenvalue
    optimizer_result = self.optimizer.minimize(
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\...\venv\Lib\site-packages\qiskit_algorithms\optimizers\gsls.py", line 125, in minimize
    x, fun, nfev, _ = self.ls_optimize(x0.size, fun, x0, var_lb, var_ub)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\...\venv\Lib\site-packages\qiskit_algorithms\optimizers\gsls.py", line 185, in ls_optimize
    directions, sample_set_x = self.sample_set(n, x, var_lb, var_ub, sample_set_size)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\...\venv\Lib\site-packages\qiskit_algorithms\optimizers\gsls.py", line 298, in sample_set
    if (points >= var_lb).all() and (points <= var_ub).all():
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: '>=' not supported between instances of 'float' and 'NoneType'
```

What should happen?

I would expect the optimization process to finish without crashing and to print the final result.

Any suggestions?

No response



Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Fix GSLs optimizer bounds if entries have None](#)
qiskit-community/qiskit-algorithms

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```
File "C:\...\venv\Lib\site-packages\qiskit_algorithms\optimizers\gsls.py", line 298, in sample_set
    if (points >= var_lb).all() and (points <= var_ub).all():
        ^^^^^^^^^^^^^^^^^^^^^
TypeError: '>=' not supported between instances of 'float' and 'NoneType'
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Any suggestions?

No response



dleidreiter added **bug** on Oct 23, 2023

woodsp-ibm mentioned this on Oct 23, 2023

[Fix GSLS optimizer bounds if entries have None #95](#)



woodsp-ibm on Oct 23, 2023

Member ...

Just an FYI having run your sample to check it now runs with the fix - Qiskit Aer has its own primitives https://qiskit.org/ecosystem/aer/apidocs/aer_primitives.html which are more efficient/better to use than using the BackendSampler/Estimator that were more intended for providers that did not have primitives for their backends.



dleidreiter on Oct 24, 2023

Author ...

Ok, thank you. :)



ElePT closed this as **completed** in [#95](#) on Oct 30, 2023

Fix GSLS optimizer bounds if entries have None #95

<> Code

Merged ElePT merged 1 commit into qiskit-community:main from woodsp-ibm:gsls_bounds on Oct 30, 2023

Conversation 2 Commits 1 Checks 0 Files changed 3

+15 -2



woodsp-ibm commented on Oct 23, 2023

Member

Summary

Fixes #94

Details and comments

When the optimizer creates its upper and lower bound arrays, any None entrys are replaced by infinity (i.e. No bound)



Fix GSLS optimizer bounds if entries have None

1ad7a3d

woodsp-ibm requested review from Cryoris and ElePT as code owners 2 years ago



coveralls commented on Oct 23, 2023

...

Pull Request Test Coverage Report for Build 6618976410

- 2 of 2 (100.0%) changed or added relevant lines in 1 file are covered.
- No unchanged relevant lines lost coverage.
- Overall coverage increased (+0.03%) to 90.554%

Totals	coverage 91%
Change from base Build 6573888143:	0.03%
Covered Lines:	6452
Relevant Lines:	7125

♥ - Coveralls



ElePT approved these changes on Oct 30, 2023

View reviewed changes

ElePT left a comment

Collaborator

LGTM, thanks!



Reviewers

ElePT



Cryoris



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

GSLS optimizer crashing with QAQA default ...

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